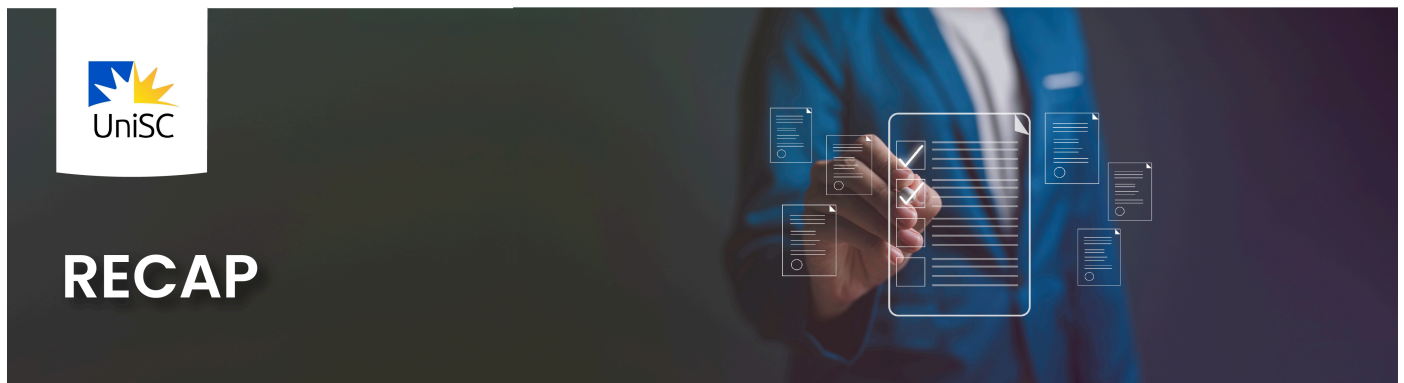


Module 4 Summary



What You Learned in Module 4

<https://learn.usc.edu.au/courses/35359/files/2918938?wrap=1>

Module 4 looks at the CAPM, which remains the workhorse of modern finance. We begin this module by looking at the key motivations underlying CAPM, and how its key parameter beta is estimated and interpreted. This coincides with the following course learning outcomes:

- **CLO 1:** Explain key concepts and theoretical foundations of financial analytics and their role in strategic decision-making.
- **CLO 2:** Apply quantitative and time-series techniques to model risk, return, and financial performance in real-world contexts.

Module 4 then moves to discussing the concept of WACC, its various components, the role of the CAPM in it, and how the concept of WACC is used in the industry. This supports the following course learning outcomes:

- **CLO 1:** Explain key concepts and theoretical foundations of financial analytics and their role in strategic decision-making.
- **CLO 2:** Apply quantitative and time-series techniques to model risk, return, and financial performance in real-world contexts.

Module 4 then explores how the graphical tool of SML is used in the investment industry to communicate essential insights related to spotting under-and overvalued assets. This supports the following course learning outcomes:

- **CLO 1:** Explain key concepts and theoretical foundations of financial analytics and their role in strategic decision-making.
- **CLO 4:** Create and communicate financial insights using effective visual, written, and oral formats suited to both technical and non-technical stakeholders.

We concluded by looking at the key empirical shortcomings of the CAPM. As we wrap up this module, you:

- 1) Understand the intuition and the basic structure of the CAPM. Know how to estimate beta.
 - 2) Understand how the CAPM is used in the industry to estimate the cost of capital. Explain how this matters for capital budgeting and project evaluation.
 - 3) Can explain the concept of the security market line (SML). Use the SML to identify undervalued and overvalued stocks.
 - 4) Understand some the limitations of the CAPM: the empirical evidence accumulated over the years casts doubts on its validity in its original form; however, the model remains useful and is widely used in the industry.
-

Practice Tasks

Practice Task 1: (Utilize the R code provided in the module to complete this exercise)

Choose a publicly traded stock of your choice (any company you're interested in, provided at least 5 years of monthly price history is available). Using R, estimate that stock's beta relative to the S&P 500 and interpret your results.

Download the data. Using `tq_get()`, download 5 years of monthly adjusted price data for your chosen stock and for the S&P 500 (ticker `"^GSPC"`).

Calculate returns. Compute monthly returns for both series.

Calculate excess returns. Using the 10-year US Treasury yield as your risk-free rate proxy (state the rate you used and the date you sourced it from), convert it to a monthly rate and subtract it from both your stock's and the market's monthly returns.

Run the regression. Estimate beta using `lm(stock_excess ~ market_excess)`.

Report your results. State the estimated beta, alpha, and R-squared from `summary()`.

Interpret your results, addressing all of the following:

- Is your stock more or less volatile than the market? What does the beta value tell you about how you'd expect this stock to behave in a market downturn versus a market rally?
- Is alpha statistically significant? What would it mean if it were positive and significant?
- What does the R-squared tell you about how much of your stock's movement is explained by the market, versus firm-specific factors?

Apply CAPM. Using your estimated beta, a risk-free rate of your choice (state it), and a market risk premium of 7%, calculate your stock's CAPM-expected return.

Would you consider this stock's estimated required return to be low-risk or high-risk relative to the market? Does this match your intuition about the company and its industry? Note one limitation of using a 5-year regression window to estimate beta.

Practice Task 2:

Consider a bond with a 6% annual coupon and a face value of \$1,000. Complete the following table. What relationships do you observe between years to maturity, yield to maturity, and the current price?

Years to Maturity	Yield to Maturity	Current Price
2	4%	
2	6%	
3	6%	
5	4%	
5	8%	

Answer available [here \(https://learn.usc.edu.au/courses/35359/files/3294280?wrap=1\)](https://learn.usc.edu.au/courses/35359/files/3294280?wrap=1).

Workshop

[Tasks \(https://learn.usc.edu.au/courses/35359/files/3282922?wrap=1\)](https://learn.usc.edu.au/courses/35359/files/3282922?wrap=1)